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Optical Property Measurements
using
the Inverse Adding-Doubling Program

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March 1995

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Abstract

This document describes a method for measuring the optical properties of scattering and absorbing materials. The general idea is that you, the budding (or budded?) scientist, make careful measurements of observables like the total reflection and transmission by the sample. These observations are then fed into a program (the inverse adding-doubling IAD program) to extract the intrinsic optical parameters for the sample. The program does this by repeatedly guessing the optical properties and comparing the expected observables with those that you have made. Upon matching, voilà, those optical properties used to generate the expected observables are the ones that characterize your sample.

If only it were that simple. I have written a couple of papers on the details of the program [1, 2], on optical properties in general [3], and on making those measurements with integrating spheres [4, 5]. Unfortunately, not one of these brings all the pieces together. Partly, this is my fault, partly it is because none of the articles is really concerned with dotting all the *i*'s and crossing all the *t*'s, partly it is because an appropriate forum for this information just doesn't exist other than in a manual or book.

This is not an amazing document. Hopefully it will get better with time. I can't spend any more time on this document at the moment. If you have questions or comments fax me the relevant information and I will reply as quickly as I can. I find it too hard to provide support over the telephone. If I get any feedback then I will try to improve things and add features that people request. Being somewhat pessimistic, I don't think that anyone will bother. Go ahead, make my day, prove me wrong.

History and Acknowledgments

The IAD program started its life as an inverse diffusion program.* Steve Jacques and I just wanted a way to find the optical properties of tissues. We published a paper ([1]) and thought that was that. We subsequently discovered that the diffusion model can generate negative reflectances for samples with large anisotropies ($g=0.9$)! Steve encouraged me to improve the inverse diffusion program. I subsequently improved that program in two important ways. First, I used the delta-Eddington approximation to eliminate the problem with the negative reflectances. Second, I replaced my simple-minded iteration method with the simplex method of Nelder and Weaver. In this manner the inverse delta-Eddington model was created. I used this in/for my thesis. During this time I was led by A. J. Welch and the cash came from some Navy contract.

About the time I was giving my final defense I realized that I could not recommend my inverse program to anyone because I had no idea how accurate it was. I started working on a paper to estimate the errors but it seemed pretty pointless. I decided to take the adding-doubling program that I had just written and substitute it for the delta-Eddington program in the inverse program. This led to the inverse adding-doubling (IAD) program. This was done in Amsterdam under the auspices of Martin van Gemert and guilders from the Fundamenteel Onderzoek voor Materie. This program was much more accurate. In fact you can make the adding-doubling method as accurate as you want by increasing the number of quadrature points. There were a couple of major snags—most notably the problem with internal reflection. It nearly killed me, but the solution was surprisingly simple as long as you understand/implement the integration by quadrature carefully.

Since that time most of the difficulties with the IAD program have been traced to exactly how the first guess is made. I spent a week working in Houston with Steve trying to solve this problem. After a bit of fussing (a lot actually) over the starting values and adding the delta-Eddington program back in (when $N = 2$) I had almost arrived at the core of the current IAD program. The last little bit of work was done during a couple of months spent at the University of Utah Laser Institute. Here the starting routine was completely rewritten. I developed the heuristics for calculating starting values that are given in the adding-doubling paper [2]. Large portions of the code were split up into smaller files and the first version of this documentation was written.

While I worked at Wellman as a post-doc, I fixed a number of minor bugs and changed the headers for the data files dramatically to bring the sphere corrections into accordance with the integrating sphere paper by Pickering and others. I also implemented the use of a stored grid to extract values from. This can yield immensely faster convergence if the optical depth or anisotropy is known *a priori*. However, if neither of these is the case, then using a stored grid does not help much.

*This is somewhat ironic because when I translated the program into ANSI-C, I did not convert the diffusion code. Version 1.4 of the IAD program does not allow one to make calculations using the diffusion model.

While at the Oregon Medical Laser Center, I converted the entire program to ANSI-C. I improved many things so that the program is much faster and more reliable. I adapted the stored-grid approach so that it works reasonably reliably. I used the **CWEB** system to improve the documentation. This work was supported by the Collins foundation.

The latest stage in the life of this program has been due to my most excellent interaction with David Royston at the FDA. He did many careful experiments and then kept asking what was going wrong. I eventually tracked many of these problems down. Often they were the result of bugs that either had crept into the C port of the code, or they were the result of coding ineptitude on my part from the very beginning. Dave also provided the incentive to incorporate absorbing slides that absorb into the calculation.

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variable	description	units
a	single scattering albedo	—
$p(\cos \theta)$	phase function	sr^{-1}
μ_s	scattering coefficient	mm^{-1}
μ'_s	reduced scattering coefficient	mm^{-1}
μ_a	absorption coefficient	mm^{-1}
τ	optical thickness	—
τ'	reduced optical thickness	—
A	area of the sphere	cm^2
a	area normalized to the total sphere area	—
r	reflection for diffuse irradiance	—
d	(subscript) detector	—
w	(subscript) walls of the sphere not including holes	—
h	(subscript) holes in sphere besides detector and sample hole	—
s	(subscript) sample	—
$(\#)$	(superscript) number of times light has been reflected by the walls of the sphere	—
c	(superscript) collimated to diffuse	—
cc	(superscript) collimated to collimated	—

Nomenclature

And many others. No time to list them all now.

Chapter 1

Introduction

Inverse adding-doubling is a method for generating the optical properties of turbid media. It converts reflection and transmission measurements of a sample (typically with an integrating sphere) to the optical properties of the sample (albedo, optical thickness, and anisotropy).

To use the IAD program, create a data file, say `xxx.dat` with a your experimental results and tack on a header describing your experiment. Drop the data file on the IAD application. You will get back a file `xxx.dat.out` that contains a set of optical properties—the albedo, the optical thickness, and the anisotropy that correspond to each measurement set. Import this file into your favorite spreadsheet and use the physical thickness of the sample to calculate absorption and scattering coefficients. (see the formulas below if you need to).

The IAD program has parameters that can be modified to allow for several different experimental geometries. Either collimated or diffuse irradiance is permitted. One, two, or three measurements can be used to determine on, two, or three optical properties. This program can be used interactively or in batch operation.

The IAD program finds the optical properties characterizing a sample by using reflection and transmission measurements. A set of optical properties is guessed and the reflection and transmission is calculated. These values are compared with the measured reflection and transmission values. If they match then the optical properties for the sample have been found. If they do not match then a new set of optical properties is guessed and the process is repeated. As van de Hulst pointed out, the approach is fairly brute force. Oh, well it works.

The adding-doubling method is used to calculate reflection and transmission. Since the adding-doubling method is an accurate solution of the radiative transport equation for all albedos, all optical depths, and all phase functions this method can be applied to any medium for which the radiative transport equation is valid. However, a number of restrictions apply. In particular the following assumptions are made

- distribution of light is independent of time,

- samples have homogeneous optical properties,
- tissue geometry is an infinite plane-parallel slab of finite thickness,
- tissue has a uniform index of refraction,
- boundaries are smooth,
- internal reflection is governed by Fresnel's law, and
- polarization of light may be ignored.

Inverse adding-doubling has a number of advantages and a number of disadvantages. The advantages are that it

- works for any combination of optical properties.
- takes into account all the interactions of a sample sandwiched between glass slides
- incorporates the effects of the integrating spheres on the measured values
- has a reasonable trade-off between accuracy and speed.

The disadvantages are that it

- requires one to use an integrating sphere
- does not account for light lost out the edges of the sample
- needs a Macintosh to analyze the data (but I do provide source code in ANSI-C)

Despite extensive testing of this program, you will probably find bugs. Call, mail or e-mail your complaints and accolades. Other useful items that can be obtained from me are

- A Macintosh disk containing an executable version of the inverse adding-doubling program, a text editor BBEdit, and a few sample data sets.
- A reprint of my paper inverse adding-doubling describing the theory and numerical accuracy of the technique.
- A reprint of the paper John Pickering wrote (reporting the work done by Niek van Wieringen under my supervision) on evaluating the experimental accuracy of the inverse adding-doubling technique.
- A copy of the goniometry paper by Jacques, Alter, and Prahl. This is enclosed because it implicitly gives a few alternate ways of measuring the collimated transmission through a sample and should be useful. Also, it is published in *Lasers in the Life Sciences* which is hard to get.

Possible future enhancements include

- Putting the diffusion code back in.
- Background processing.
- Testing Brent's algorithm to do the inversion.
- Future drivers for this program will include automatic correction of baseline stray light measurement for double sphere problems. (if you don't know, then you probably don't care. Basically, this issue can be avoided completely by careful experimental technique.)

1.1 Other news

There is only one executable version of the program available now. This may be used with spectrophotometric or regular applications. There are also a handful of sample data files and results included. If you are having problems compare the header section of your data files with those from the samples very carefully before complaining to me. Most problems that have been reported have been traced to a mistake in the header files.

If you want to abort a calculation in progress press apple-option-esc. Just kill the program.

The program is intended to be used in drag-and-drop mode. Just highlight all the data files that you want to process and then drag them onto the IAD application. Each file should be processed just fine.

There is now no distinction between the spectra and regular programs. The code checks the first entry in a line to see if it is less than 1. If not then it assumes that the entry is a wavelength and discards it. The next value is assumed to be the reflection.

If you are having problems check the headers of your files very carefully for potential errors. I do essentially no error checking of the headers—you're welcome to send me code to do this I just haven't time to do it.

Source code is available. Currently you must have bought the book Numerical Recipes in C by Press et al. before I can send you the code. Just assure me that you have a copy so that we do not violate copyright agreements.

The algorithm for finding the optical properties is different. It is really rather simple-minded, but it is faster and more reliable so I suppose that is what I get for trying to be clever.

Anyone who is interested in porting this code to the IBM, should contact me. It should be rather easy since it is written in ANSI-C. Only some make file should be needed.

The program does not work in the background.

Good luck and happy optical property calculating.

<code>web1.4</code>	source <code>*.w</code> files
<code>code1.4</code>	the above files converted to <code>*.c</code> files
<code>listing</code>	A complete $\text{\TeX}$ listing of the basic AD and IAD routines. Run these through plain $\text{\TeX}$ using the included <code>cwebmac.tex</code> macro package to obtain a complete cross-indexed listing of the program.
<code>nr</code>	slightly modified files from the numerical recipes package. Do not look at these if you have not bought the numerical recipes book.
<code>mac</code>	files specific to a macintosh application

1.2 Porting

I am interested in getting this beast ported to a couple other platforms, namely UNIX and the PC. Basically, the code should compile without any problem (ha ha) because I have not used any particularly special features of C on the Macintosh. Segmenting the code may be a problem, but it wasn't on the Macintosh so it should not be on the PC. On the PC, far pointers may be needed, but I don't really know.

The other detail would be to write some make files for the different systems. The files `iad.make` and `ad.make` show the make files for MPW. These files (although filled with funky characters) should show the dependencies pretty well.

The actual source files `foo.w` are written in CWEB, from which the ANSI-C source code files `foo.c` and `foo.h` files are generated. The documentation `foo.tex` is also generated from the same base `foo.w` files. This keeps the code and the documentation in one place and furthermore allows me to use the typesetting capabilities of $\text{\TeX}$ in the program comments. I highly recommend the CWEB system. It is available for anonymous ftp from `ftp.shsu.edu` in `tex-archive/web/cweb` as `cweb-3.2.tar.gz`.

In deference to those who do not want to do Literate Programming, I include the C source files in a separate directory.

I have made the names much more cryptic than they used to be, in the hopes that this would facilitate porting to the IBM platform by limiting the names to eight letters. So far no one has volunteered to try compiling these on a Unix box or a PC. Oh well.

Chapter 2

Background

2.1 Optical properties

Light propagation through tissue is characterized by radiative transport theory with three different optical properties:

- the absorption coefficient μ_a [m^{-1}],
- the scattering coefficient μ_s [m^{-1}], and
- the phase function $p(\hat{s}, \hat{s}')$ [sr^{-1}].

The reciprocal of the absorption or of the scattering coefficient ($1/\mu_a$ or $1/\mu_s$) is the average distance that light will travel before being absorbed or scattered respectively. The phase function describes the fraction of light scattered from the direction $\hat{s}$ into the direction $\hat{s}'$. The phase function is normalized so that its integral over all directions is one,

$$\int_{4\pi} p(\hat{s}, \hat{s}') d\omega' = 1.$$

Here $d\omega'$ is a differential solid angle in the $\hat{s}'$ direction. The functional form of the phase function is usually not known. In these cases the phase function is characterized by its average cosine

$$g = \int_{4\pi} p(\hat{s}, \hat{s}') (\hat{s} \cdot \hat{s}') d\omega'.$$

The average cosine of the phase function g , is usually called the anisotropy coefficient. The anisotropy coefficient varies between isotropic scattering ($g = 0$) and complete forward scattering ($g = 1$). Notice that the integrals in equations (1) and (2) are independent of the choice of $\hat{s}$. This means that the scattering profile is independent of the incoming light direction. Not surprisingly, a single anisotropy coefficient is inadequate to describe tissues that have special

directions along which light scatters better or worse. For example, tissues (e.g., tendon) that have preferential scattering along oriented collagen fibers in a tissue.

The default phase function used by this program is the Henyey-Greenstein

$$p(\cos \theta) = \frac{1}{4\pi} \frac{1 - g^2}{[1 + g^2 - 2g\mu]^{3/2}}$$

Other phase functions can be used, but currently this is the only phase function implemented.

Using these three basic optical properties, several useful parameters can be derived. The first is the total attenuation coefficient

$$\mu_t = \mu_a + \mu_s.$$

and the effective attenuation coefficient

$$\mu_{\text{eff}} = \sqrt{3\mu_a(\mu_a + \mu_s(1 - g))}$$

Often the the anisotropy coefficient is not known. In this case the reduced scattering coefficient is useful. It comes from similarity relations that reduce the number of necessary parameters from three down to just two. Two possible choices for a similarity transformation are

$$\mu'_s = (1 - g^2)\mu_s \quad \text{or} \quad \mu'_s = (1 - g)\mu_s$$

The latter is the most commonly used. The former comes from the delta-Eddington approximation to a Henyey-Greenstein phase function.

Of course, now still more parameters can be generated. The reduced total attenuation coefficient.

$$\mu'_t = \mu_a + \mu'_s$$

The effective attenuation coefficient becomes

$$\mu_{\text{eff}} = \sqrt{3\mu_a\mu'_t}$$

Two other important parameters are the albedo a and the optical distance τ defined in the next section.

2.2 Dimensionless optical properties

Light propagation is characterized by three parameters: the albedo, the optical depth or thickness, and the phase function. The albedo a is a dimensionless parameter defined as the ratio of scattering coefficient to the sum of scattering and absorption coefficients

$$a = \frac{\mu_s}{\mu_s + \mu_a}$$

The optical depth τ is the product of the tissue thickness and the sum of the scattering and absorption coefficients

$$\tau = d(\mu_s + \mu_a).$$

The parameter g is sometimes referred to as the anisotropy coefficient. The scattering coefficient μ_s and the absorption coefficient μ_a are given by

$$\mu_s = \frac{a\tau}{d} \quad \mu_a = \frac{(1-a)\tau}{d}.$$

The reduced optical albedo and optical depth are given by

$$a' = \frac{a(1-g)}{1-ag} \quad \tau' = (1-ag)\tau.$$

The inverse relations are

$$a = \frac{a'}{1-g+a'g} \quad \tau = \tau' + \frac{a'\tau'g}{1-g}.$$

and

$$\mu'_s = \frac{a'\tau'}{d} \quad \mu_a = \frac{(1-a')\tau'}{d} \quad \mu_s = \frac{a'\tau'}{(1-g)d}.$$

2.3 Integrating Spheres

There is a long history of the use of integrating spheres. Good 'ole Ulbrecht reported the first one a hundred years ago. You can make your own easily. Just get a sphere, e.g., a child's ball and cut it in half. Paint the inside with BaSO₄, glue it back together attach detectors and voilà! Or you can plunk down a thousand bucks and buy one from LabSphere.

Next you need some calibration plates. The easiest NIST (no longer NBS) traceable standards obtainable are also from LabSphere. They cost about \$200 apiece. Or you can use freshly prepared BaSO₄ plates.

OK, now you need to decide how you will actually make the measurements. The two methods are single beam and double beam methods.

If you want to use a double integrating sphere set-up then the single beam method is required. Furthermore, this is the simplest apparatus to construct. No lenses and stuff are needed. However, this method requires that you make calibration measurements on your integrating sphere. These are made by measuring the size of all the ports and making measurements with the calibration plates that you bought. See Pickering *et al.* for details [4].

Another way is to use a single sphere which has a sample and a reference port and differentially detect. This has distinct advantages when measuring very low reflection values. The reference plate can be say 10% and a factor of ten improvement in signal can be obtained if the electronics are good. This method is found in the better spectrophotometers. I am currently worrying about the corrections for such an apparatus. Currently, the best that can be done is to normalize the measured values using calibration scans.

Chapter 3

Experimental Considerations

There are a whole host of experimental problems which plague measurements of reflection and transmission. Inattention to experimental details will yield garbage. Just look at the variation in the optical properties gathered together in Cheong *et al.*[3]. These can be attributed to two sources of error—inattention to the details of the experimental measurement and to flakey optical models. Since presumably, the adding-doubling method is not flakey, you are saved from the second source of error. But I can do nothing about saving you from poor experimental technique except to warn you about various things.

3.1 Freshness

If the sample is not fresh then optical properties will differ from those measured *in vivo*. If the tissue is not in excellent condition then there will be problems. By the way, there has been essentially no work done on the changes in optical properties which take place when a tissue is removed. There has been minor work done on the changes in optical properties with heating, but nothing to that definitively says that the optical properties measured *ex vivo* are even close to those *in vitro*.

3.2 Boundaries

The need for glass or quartz slides at the boundaries is caused by the rough surface of most tissues. Since the boundary is characterized by Fresnel reflection, controlling this is critical. Water or some other fluid should be used to ensure that no bubbles are formed between the slide and the tissue. Furthermore, the water is available to fill the voids between the tissue and the glass. Since the index of refraction of glass is 1.5 and that of tissue ranges from 1.33 to say 1.45,

the boundaries are not perfectly matched. You just have to live with this or experiment with various immersion oils and various glasses and plastics achieve perfection. Of course other experimental difficulties are most certainly larger than this one.

Oh yes, the slides also help with the next two problems also.

3.3 Hydration

Optical properties definitely change with the amount of water in the tissue. It is important to store the tissue in air tight containers sandwiched between moist (with isotonic saline) towels at cool (above freezing) temperatures. If you don't believe me, then leave a piece of tissue exposed to air for 48 hours. It will appear quite different from fresh tissue. Since your eye sees reflected and transmitted light, any measurements will also differ.

3.4 Sample Thickness

There are two parts to this problem. First obtaining uniformly thick slices of tissue. The second is concerns what thickness of sample is desirable.

3.5 Sample Diameter

How big should the sample be? First, it should completely cover the port in the integrating sphere. If not, then make the port smaller. Second, you want the distance from the edge of the irradiating beam on the sample to the edge of the port (h) to be much larger than the lateral propagation distance. Thus we want

$$h \gg \frac{1}{\mu_a + \mu'_s}$$

If this is not satisfied, then light will be lost out the sides of the sample and the loss will be attributed to absorption. The absorption coefficients generated for the sample will be too large. This is particularly important with thick samples in the red and infrared regions of the spectrum.

3.6 Variability

From spot to spot on one sample and from sample to sample. Make enough measurements that you have some idea of the average value and variance for each sample as well as the variability from sample to sample.

3.7 Blood

Difficult one to call. Should samples be measured with or without blood. Clearly neither really is a good example of the in vivo situation. If samples have no blood then the optical properties of the sample itself are being measured. The contribution to blood can be added later.

3.8 Freezing

To obtain very thin sections it is necessary to freeze the sample. Rumor has it that the freezing process changes the optical properties. The evidence for this has been on muscle tissue. Other tissues like aorta and dermis are different and not so susceptible to freezing artifact.

3.9 Interference

Serious pain in the wahzoo. The interference usually comes from the glass slides which sandwich the tissue sample. When a laser beam strikes the air-glass surface it is reflected. It is also reflected at the glass-tissue surface. This reflection can interfere with the first reflection. One would expect this to be a negligible effect, but in practice it can be nearly 10%. It is easily demonstrated by placing a microscope slide in the path of a laser beam and measuring the transmission. Then move the slide slightly and observe the transmission. An even more dramatic effect is achieved when specular reflection is measured.

What can be done? Sit down with a box of microscope slides and go through them one by one observing the reflection profiles on a wall. Discard all those that show definite interference effects. These should be the least optically flat of the whole box.

Another possibility is to use very thick glass plates. By slightly rotating the plate the reflection from the second surface will be displaced sufficiently that the two beams will no longer be aligned and consequently will not interfere. Thick plates are not ideal because they displace the integrating spheres and consequently all some diffuse light to be lost before it can enter the integrating sphere.

A final possibility is to use optically flat plates. Interference will be constant over the surface and can be measured and accounted for. This is the best option, but also the most expensive.

3.10 Collimated Transmission

By collimated transmission, I mean light that has not interacted with the sample through absorption or scattering. A better name would be on-axis transmission or unscattered transmission. The phrase on-axis transmission is also confusing because the light transmitted directly through the sample is a mixture of

scattered and unscattered light. It is noteworthy that no-one has made coherence measurements through tissue samples. Some time-of-flight measurements have been made, and I suppose that the ballistic photons (those that pass right through the sample are related to those photons which retain their coherence, but it has not been studied to my knowledge.

The collimated transmission is the most difficult measurement to make. It is frequently underestimated. Be careful. Do yourself a favor and plot your measured values for collimated transmission versus the concentration or sample thickness. If the line is not linear then you have problems. Furthermore, you should check to make sure that the data passes through the correct value when concentration or sample thickness is zero. This is easily calculated using the indices of refraction. For example, for water between glass slides the transmission should be

$$T_c = (0.96)(0.996)(0.996)(0.96) = 0.914.$$

where I have neglected to include extra internal reflections. The IAD program does, but then it is good at that sort of thing.

Next, do a quick calculation to ensure that the amount of scattered light reaching the detector is significantly less than the estimated value for the collimated transmission. Since $T_c \propto \exp(-\mu_t d)$ it gets small, very quickly! Do not underestimate the enormity of the smallness of T_c !

Remember, if your collimated transmission data does not pass the above test, then the output from the IAD program must be suspect. *Garbage in, garbage out.*

See the entire chapter below devoted to this annoying little problem.

3.11 Polarization

Very little work has been done in this area. Svasaand published a bit of data several years ago. Not much since then. This particular implementation of the adding-doubling program treats everything as unpolarized light. If you want to include polarization—go for it.

3.12 Difficulties

When the program does not converge...what should I do? There are two possibilities. First there is something wrong with the data. Look at it very carefully. Is the reflection smaller than it possibly could be? Does the total reflection plus the total transmission equal a larger value than one?

The second possibility is that the program got stuck in a very narrow valley in the function that it was trying to minimize. The best solution is to specify the starting value yourself. The program does the best it can, but guessing the starting value is difficult because it is exactly the problem you are trying to solve. Anyway, specifying your own starting value can be done with two lines of code. The details are given below in the implementation section.

Chapter 4

The program

4.1 The file structure

When a lot of data must be processed, or when just one point is needed, a data file must be created with the appropriate header. This header describes properties that are constant for all the measurements in the data file. See the questions section for a description of those funny V% symbols.

line	info	description
1	1	version number...must be 1
2	description	line ignored
3	description	line ignored
4	nslab, ntop, nbottom	indices of refraction
5	irrad	irrad = 0 means collimated irradiance irrad <> 0 means diffuse irradiance
6	spec refl	spec refl = 0 => V1% = scattered refl. (total - unscattered) spec refl <> 0 => V1% = total refl. (scat + unscattered)
7	coll trans	coll trans = 0 => V2% = scattered trans. (total - unscattered) coll trans <> 0 => V2% = total trans. (scat + unscattered)
8	anisotropy	gvalue = default g when V3%=0.
9	spheres	spheres = 0 for none or unknown spheres spheres = 1 if only one sphere was used spheres = 2 if double spheres were used (If no spheres or spheres with unknown calibration coefficients were used then set spheres = 0)
10	# of measurements	measures = 1 then file contains only V1% (V2% and V3% assumed zero)

```

measures = 2 then file has V1% and V2%
                (V3% assumed zero)
measures = 3 then file has V1%, V2%, V3%
11 b1_r,b2_r,s/A_r,a_r,m_r first sphere constants
                        ignored if spheres=0
12 b1_t,b2_t,s/A_t,a_t,m_r second sphere constants
                        ignored if spheres=0 or spheres=1
13 nfluges                2,4,6,8,...number of fluxes, usually 4.
14 description            ignored

```

4.2 Example 1

For example, if you have made measurements on a piece of tissue which was sandwiched between glass slides, and you used a laser beam as your source, and you arranged it so that the specular reflection from the front surface of the glass slide left the integrating sphere when the reflection measurement was made, and you collected all the light in the transmission measurement, and you did not make collimated transmission measurements, and you had a good idea that the anisotropy of the tissue was 0.8 at the wavelength you were using and you had no idea what the calibration factors for your integrating spheres were, then this is the sort of header that you would use for your data file is

```

1                {inverse adding-doubling version number}
Sample header    {ignored}
by Scott Prah1   {also ignored}
1.4 1.52 1.52    {tissue, glass above and below}
0                {collimated irradiance}
0                {Specular reflection is not included in V1%}
1                {Collimated transmission is included in V2%}
0.8              {Default anisotropy value}
0                {No sphere coefficients}
2                {Number of measurements}
                {ignored -- would be refl sphere coef}
                {ignored -- would be trans sphere coef}
4                {four fluxes}
Rd      Tt      {ignored}

```

4.3 Example 2

For a version 2 file, you can specify if a baffle was present or not by including a zero or one.

```

2                {inverse adding-doubling version number}
Sample header    {ignored}
by Scott Prah1   {also ignored}

```

```

1.4 1.52 1.52          {tissue, glass above and below}
0                      {collimated irradiance}
0                      {Specular reflection is not included in V1%}
1                      {Collimated transmission is included in V2%}
0.8                    {Default anisotropy value}
2                      {Two sphere coefficients}
2                      {Number of measurements}
b1 b2 s/A alpha m baffle {Refl sphere coef baffle=0 if no baffle}
b1 b2 s/A alpha m      {Trans sphere coef}
4                      {number of quadrature points}
Rd   Tt                {ignored}

```

Note that the inverse process will use four fluxes. This is a reasonable number and probably the only really workable number. More fluxes takes a long time.

After the header comes your data. In this case it would be two columns. The first column would be the diffuse reflection values and the second column would be the total transmission values. The should always be between zero and one. The sum must be less than one.

4.4 How the header info is used

Based upon the input data different assumptions will be made. Ideally one would always have three measurements. Unfortunately, often the collimated transmission measurement is not measurable. This could happen with a piece of tissue only a millimeter thick or so. Furthermore, sometimes $V_1\%$ is zero also. In this case only one parameter can be found.

- One measurement—
 1. The optical thickness of the slab is assumed infinite and the anisotropy coefficient set to the default value. The albedo is varied until the correct value for the reflection ($V_1\%$) is obtained.
- Two measurements—
 1. If $V_2\%$ is non-zero then the anisotropy coefficient set to the default value. The albedo and optical depth are varied until the correct values of $V_1\%$ and $V_2\%$ are obtained. If $V_2\%$ is zero then the optical depth is assumed infinite and only the albedo is varied.
 2. If $V_2\%$ is zero then the optical thickness of the slab is assumed infinite and the anisotropy coefficient is set to the default value. The albedo is varied until the correct value for the reflection ($V_1\%$) is obtained.
- Three measurements—
 1. If both $V_2\%$ and $V_3\%$ are non-zero, then $V_3\%$ is used to calculate the optical thickness. The albedo and anisotropy are varied until $V_1\%$ and $V_2\%$ are matched.

2. If $V_3\%$ is zero but not $V_2\%$, then anisotropy is fixed at the default value and the albedo and optical depth are varied until the correct values for $V_1\%$ and $V_2\%$ are obtained.
3. If both $V_2\%$ and $V_3\%$ are zero then the optical thickness of the slab is assumed infinite and the anisotropy coefficient is set to the default value. The albedo is varied until the correct value for the reflection ($V_1\%$) is obtained.

Chapter 5

Beckman UV 5270 Spectrophotometer

5.1 Introduction

This memo* is intended to describe how reflection and transmission experiments might be made with the Beckman UV 5270. Particular attention is paid to the idiosyncrasies and limitations of the Beckman.

First things first. The wavelength is not what it might seem and it should be understood that the number displayed by the Beckman is not the number recorded by the PC attached to the Beckman. For simplicity we † choose not to use the digital signal generated by the Beckman, but rather just took the analog signal out of the back and ran it through an A/D board. The output from the A/D board is recorded by the computer.

The measurement need to be divided into four wavelength regions. The UV (250–400 nm), the visible (400–800 nm), the near-infrared (800–1300 nm), and the less-near-infrared (1300–2500 nm). Each will be discussed in turn and skin will be assumed as the sample. However, before we do this some remarks which are applicable regardless of wavelength should be mentioned.

5.2 General

- The Beckman should be turned on into the warm-up mode and left there for about 10–15 minutes before turning it on completely. The tungsten lamp is used exclusively except in the UV when the deuterium lamp is needed for extra photo ummph.

*Vasan Venugopalan at Wellman helped revise this section. So he shares the blame and the glory.

†Steve Jacques and I spent a week taking data with the Beckman back in 1986. Steve found the A/D converter and I hacked a BASIC program together that week just to get data. This is the pretty much the system that you have now.

- For reflectance measurements, the skin sample is placed against a quartz or glass plate and water was used to match the refractive index of the skin with this plate. Oil was not used[‡]. Before performing all measurements a 0% and 99% reflectance scan must be done. Also when doing this scan the 0% and 100% knobs must be adjusted so that the Beckman never reads below 0% or above 100% for either of these scans over the whole wavelength region. Note that the Spectralon reflectance standards should never be put in contact with oil as it renders them defective (they absorb oil, but are hydrophobic). When doing the reference scans no matching fluid or glass/quartz plate is used.
- When performing the transmission measurements, the sample is placed between two glass or quartz or glass plates. This sample is placed in front of the sample beam which enters the sphere. In this case a BaSO₄ plate is placed in both the sample and reference ports. The 0% and 100% transmission measurements are performed by placing either an opaque or no sample in the path of the beam. When doing these scans the 0% and 100% knobs must be adjusted so that the Beckman never reads below 0% or above 100% for either of these scans over the whole wavelength region. For the tissue samples, water must also be included to keep the tissue sample hydrated. However when testing epidermis and dermis make sure that the amount of water added does not artificially increase the thickness of the sample. Of course for the transmission measurements the thickness of the sample must be measured. This is done by measuring the thickness of the quartz/glass plates used with and without the sample inserted in between. You need to make several measurements in different places because most samples are not very uniform in thickness.
- These reference scans should be performed three additional times (e.g., once just before lunch, once after lunch and definitely once just before shutting the machine down for the day.) It is also advisable to scan the 10% or 20% standards at this time if they are to be used for the days measurements. The times when this is necessary will become clearer as you read the rest of this memo.

5.3 The Ultraviolet

There are a few problems associated with this region.

- First, the Beckman illuminates with only a single wavelength and detects all light falling on the detector in the integrating sphere. This means that if light is absorbed and the generates fluorescence, much more light will be detected for that wavelength than should be. This was pointed out

[‡]Why the worry about oil and water, well if you are interested in hydration effects then it could be an issue. Really, I only considered oil because it has been used on psoriatic lesions to reduce backscattering from the flakey outer layers

by Rox, in his (in)famous skin optics paper [7]. Rox used a solar-blind photomultiplier tube (Hamamatsu? R456) to prevent detecting visible fluorescence. Thus the same photomultiplier should be used when making the measurements in the wavelength region 250–300 nm. (It should be in one of the drawers in the Beckman room.) Ideally these measurements should also be made with some sort of fluorimeter with the excitation and emission monochromators scanning at the identical speed and wavelength.

- The reflected and transmitted light can be very small (especially for darker skin types). The measurements are correspondingly noisy. We used a 20% reflection plate as the reference target and adjust the 0% and 100% dials on the Beckman to effectively utilize the full range of the machine. As a result 0–100% will then correspond to a reflectance between 0 and 20%. This improves the generated signal significantly. It might be that the electronics in the Beckman just don't work so well for very small voltages.
- Because glass transmits poorly in this region, quartz plates must be used as the window against which the sample is placed for the reflection measurement.
- Probably most troubling is that in the ultraviolet region the amount of light falling on the detector is necessarily small. The Beckman opens and closes the slits on the monochromator to ensure that the signal stays reasonably constant. This ensures good signal. Unfortunately, it also increases the bandpass. Watch the dial below the wavelength indicator to get some idea of how large the bandpass is during a typical scan. You might be surprised.
- Of course this is where you need to use the deuterium lamp.

5.4 The Visible and Near Infrared (400–1300 nm)

- There is quite a lot of light in this region. The detector begins to fail at around 800 nm (or was it 700 nm?). Anyway the spectrophotometer must be adjusted for infra-red operation. This means that the machine must be turned off, and the setting on the sphere accessory (the lever on the back of the spectrophotometer behind the sphere) switched to IR mono sphere as well as some dinky little switch near the wavelength indicator.
- Once the machine is working in the infrared, you should be aware that the wavelengths now change four times as fast as in the visible. The bandpass indicator is likewise increased in magnitude.
- Also once we are in the infrared you must specify the higher wavelength as the starting wavelength λ_s . The ending wavelength λ_e must be

$$\lambda_e = \lambda_s - \frac{\lambda_s - \lambda_e}{4}$$

because the computer has no idea that the Beckman is operating in the infrared. For example if you want to scan from 800 nm to 2500 nm, You set the dial to 2500 nm and specify the starting wavelength as 2500 nm and the ending wavelength as 2075 nm.

- The scan speed is automatically increased by a factor of 4 once we switch into the IR. Thus scan speed should reduced by depressing the 1/4 nm/s switch from the 4/16 nm/s switch to keep the scan speed at 4 nm/s in the IR.

5.5 The Near Infrared (1300–2500 nm)

- The reflection of skin is very poor in this region. The strong absorption by water in skin is comparable to or exceeds the scattering coefficient. Anyway there is not a whole lot of signal.
- The solution once again is to replace the reference plate with a 10 or 20% Spectralon standard. The experiments proceed as for the ultraviolet. The absorption by glass is still pretty small and quartz need not be used.
- The 0% reflection measurements are a problem. It turns out that the back of the sphere door reflects significant amounts of light in this wavelength range. I remove the door completely and turn off the room lights and sneak off during these calibration scans. All the scans are done in a dark room.

Chapter 6

Beam and sample size

This is excerpted from Niek Van Wieringen’s report *The limitations of the determination of optical properties of tissue with a double integrating sphere set-up* (1990). This describes an experiment of some relevance to the issue of selecting a beam size and an integrating sphere port size. It has a few problems, most notably in how the total light measurement was calculated, but the original data is gone so it can’t be fixed.

6.1 Introduction

One problem associated with making measurements of the optical properties of materials, is light lost out the sides of the sample holder. Lost light leads to erroneously high absorption coefficients because all lost light is attributed to absorption by the sample. The total amount of light collected is the sum of:

- the light specularly reflected out of the reflectance sphere,
- the light collected by the reflectance sphere,
- the light collected by the transmission sphere, and
- the light transmitted unscattered out of the transmission sphere.

As light is scattered by the sample some may leave the holder laterally (figure 6.1). The amount of light is be lost will depend on the thickness d of the sample, the radius of the holes in the spheres R , and the radius of the beam r .

6.2 Method

Reflection and transmission measurements with a double integrating sphere set-up were used to find the total collected light for non-absorbing samples with various physical thicknesses d , scattering coefficients μ_s , and optical thicknesses $\tau = \mu_s d$. All experiments were done with a 1 mm HeNe laser and various

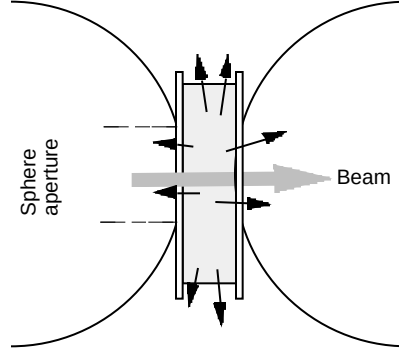


Figure 6.1: Light losses via the sides of the sample holder. The dashed lines indicate the diameter of the hole in the integrating sphere.

R (mm)	d (mm)	$R - r$ (mm)	$\frac{\tau_r}{\tau} = \frac{R - r}{d}$
10	1.95	4.5	2.3
10	0.50	4.5	9.0
19	1.95	9.0	4.6
25	1.00	12.0	12.0

Table 6.1: Sizes of the holes in the spheres R and the thickness of the sample d . In all experiments the beam diameter was 1 mm ($r = 0.5$ mm). The last column is the ratio of the radial and axial optical distances.

concentrations of Intralipid-10% to create sample optical thicknesses that varied from zero to about 55. The sizes of sphere holes and physical thicknesses are shown in table 6.1.

Experimentally, the measurement of f_0 can be eliminated by orienting the sample or the incident beam in such a way that the specular reflection does not leave the reflection sphere. The need to measure f_3 likewise can be avoided by plugging the on-axis port of the transmission sphere with a reference target with the same reflectance as the sphere (actually 100% would be good to.) This was not done because we wanted to assess how much light was lost in experiments that used the collimated transmission measurement.

6.3 Theory

The straightforward way of figuring out the total light collected would be to add the normalized outputs from the reflection, diffuse transmission, and unscattered detectors. For example, the expression would be

$$V\% = \frac{V - V_0}{V_{100\%} - V_0}$$

Where V_0 is the measurement with the laser off and $V_{100\%}$ the measurement when the laser light is directly incident on the detector. The problem with this approach is that the intrinsic absorption by the sphere walls is ignored as are the losses from other ports in the spheres—these should be taken into account using the formulas for a double integrating sphere apparatus. This is what Niek did, but he used the formula for a single integrating sphere to correct his numbers. He also used a slightly incorrect form of this equation and therefore his results will be slightly off (I would guess no more than 5%). Rather than repeat an incorrect derivation, I'll just skip to the results.

There is an interesting unresolved problem here. Is there a simplification possible in the full double integrating sphere formulas for the total light collected? How should the analysis of the data be done properly?

6.4 Results

The total light collected was measured for different combinations of the sphere aperture and the thickness of the sample d . For each combination the scattering coefficient was varied by changing the concentration of intralipid. The results of these experiments are shown in figure 6.2.

6.5 Discussion

The total light collected has roughly the same dependence on the optical thickness for the three smaller port configurations (figure 6.2).

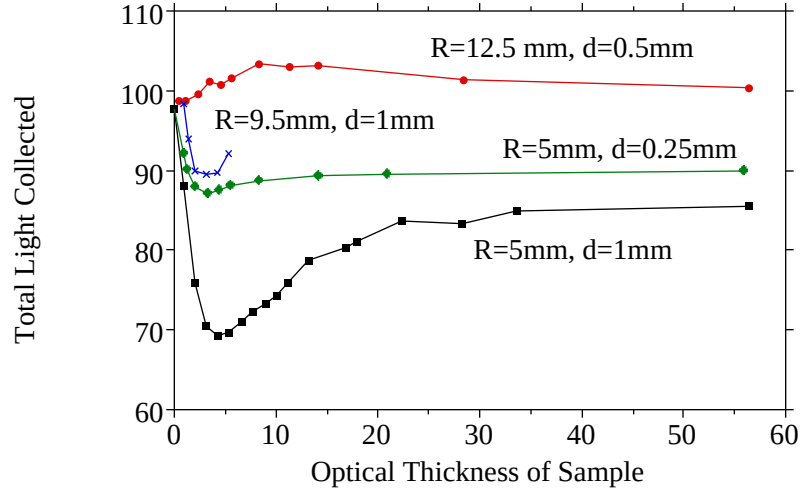


Figure 6.2: The total collected light versus the optical distance through the sample τ . Here d is the physical thickness of the sample, R is the radius of the port in the integrating sphere, and $\tau = \mu_s d$. The beam diameter was 1 mm .

1. When $\tau = 0$ there is no scattering and therefore no light is lost. (The total amount of light does not equal 100% because the specular reflection from the sample was not collected.)
2. As τ begins to increase more light is scattered and therefore more light is lost until a maximum value is no longer collected.
3. As τ continues to increase so does the optical distance in the lateral or radial direction (from the side of the laser beam to the edge of the sample aperture)

$$\tau_r = \mu_s(R - r) = \tau \frac{(R - r)}{d}$$

Therefore, the optical thickness eventually increases to the point that the number of scattering events encountered by a photons is so large that they travel only a short distance laterally and are no longer lost ($\tau_r \gg \tau$).

4. Finally, as $\tau \rightarrow \infty$ (for fixed d), causes the radial optical distance to become so large that no light escapes laterally and the total collected light increases asymptotically to a fixed value.

Consider the curves for a sphere aperture of $R = 5$ mm . The maximum discrepancy occurs when $\tau \approx 4$. When the sample thickness is 2 mm , the total collected light is always less than when the sample thickness is 0.5 mm . The relative radial path lengths are nearly a factor of four greater in the 2 mm case, and photons will encounter four times more scattering events before they might

be lost. Finally we note that the limiting values for total collected light do not differ significantly as $\tau_d \rightarrow \infty$.

When the sphere aperture is doubled in size $h = 19$ mm, then the total light collected is increased for all values of τ . Summarizing, it appears that:

- the overall collection efficiency is determined primarily by τ_r/τ , and
- the limiting value of for the total light collected as $\tau_d \rightarrow \infty$ is primarily determined by the size of the hole in the sphere.

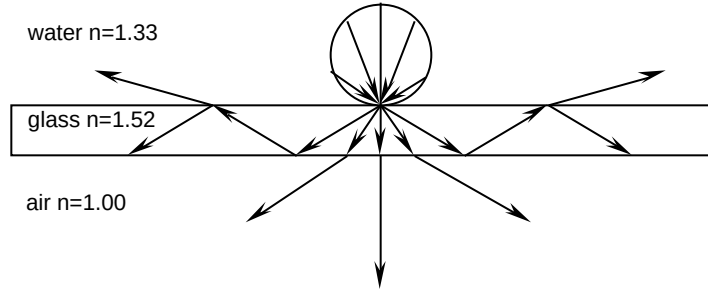


Figure 6.3: A glass slide with light incident from all possible angles from the water side. Since there is no critical angle for the water-air transition most light will not be reflected by this boundary. However, total internal reflection will occur in the glass slide for all light incident at angles greater than 40.5° . This corresponds to angles in the water of greater than 48.8° . Thus any light incident from the water at angles larger than this stand a chance of “walking” down the inside of the glass slide.

The second point bears closer examination. In short, the argument is because τ_r is always larger than τ and therefore as $\tau \rightarrow \infty$ it may be expected that no light can escape via the sides of the sample holder. However light does escape and a mechanism by which this can happen is via the glass slides of the holder, as in figure 6.3. The circle on the top represents light incident at all angles on a point on the liquid/glass boundary. The light that passes this boundary makes angles of at most 59.7° with the normal of the slide (represented by the cone inside the glass slide). At the glass-air boundary, light incident at angles larger than the critical angle $\theta_c = 40.49^\circ$ will be reflected totally and otherwise partially. Also at the glass-liquid boundary light will be reflected. If light has been reflected inside the glass slide, a couple of times it will have travelled some distance in the radial direction. For small test apertures the possibility arises that light is conducted sideways and escapes.

This possibility is tested with the following experiment. A thick piece of white perspex, strongly scattering and weakly absorbing, is placed in front of the test aperture of a single sphere set up and percentage of light reflected is measured. This measurement is repeated, this time however with a glass slide against the perspex (‘glued’ together with a very thin layer of water in order to

	spot radius	$R = 5 \text{ mm}$	$R = 12.5 \text{ mm}$
without glass	$5 \pm 1 \text{ mm}$	94.3%	94.3%
with glass	$12 \pm 2 \text{ mm}$	83.5%	90.7%

Table 6.2: The fraction of light collected using a perspex plate with and without a glass slide in front. The spot radius is the size of the glow ball on the perspex. Both experiments used the same beam diameter of 1 mm .

minimize specular reflection). The experiment is done for two different sizes of the diameter of the sample aperture R . The percentage light that is reflected, is shown in table 6.2.

The measurements with and without glass slides should not be compared with each other because it more light may have been transmitted in the second case. However, the results for different sample apertures are comparable and more light is lost when a glass slide is present. The glass slide ‘conducts’ light to the sides of the sample holder.

This is substantiated by another simple experiment that consists of measuring the diameter of the glow ball of scattered light in the perspex plate. When a glass slide is present the glow ball is more than twice the diameter measured with the perspex alone (table 6.2) This increase of the glow ball also appears when Intralipid is used instead of the perspex plate. Thus more light can be recovered by using a larger sample aperture. This effect is present in figure 6.1, where the largest sphere aperture gathered the most light. However, for this combination the total percentage of collected light exceeds 100% when τ is about 10. This of course is physically impossible, but it has to be taken into account that an approximation is used to determine the percentage of reflected and diffusely transmitted light. Still, this approximation is used in all four combinations and therefore their relation to one another stands.

Chapter 7

Collimated Measurements

Some thoughts on collimated transmission measurements that I had a long time ago.

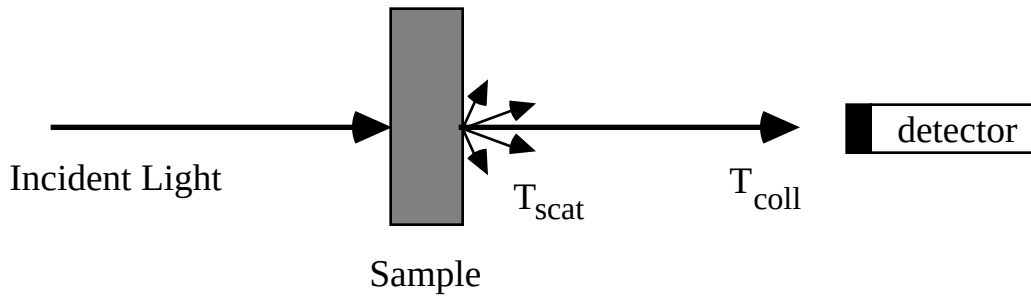


Figure 7.1: Diagrams showing how the reference measurements are made.

By far the most common measurement in the measurement of optical properties is the attenuation of collimated light through a tissue. For low albedo media, this is a simple process. The ratio of the light transmitted through a sample to the light incident is the collimated transmission. Beer's law permits computation of the total attenuation coefficient (or in the case of very low albedos the absorption coefficient)

$$\mu_a = -\ln(T_{\text{coll}})$$

This measurement need not be done with a wide beam since attenuation is uniform over beam's profile. However, the sample must be uniform in thickness and composition. An the detector cannot be moved.

For scattering tissues, measurement of primary transmission is more complicated. The experiment is the same—transmission of collimated light through a sample is measured by taking the ratio of the collimated light passing through

the sample to the total amount of light incident on the sample. The general idea is to use

$$\mu_a + \mu_s = -\ln(T_{\text{coll}})$$

to obtain the total attenuation coefficient.

It should be emphasized at this point that there does not exist a physical measurement which will give the reduced total attenuation coefficient

$$\mu'_t = \mu_a + \mu'_s = ?$$

Don't forget this, and everyone will be happier.

Unfortunately, if the sample scatters light then any measurement consists of contributions from both primary (unscattered or collimated) and secondary (scattered) light.

$$T_{\text{measured}} = T_{\text{scattered}} + T_{\text{unscattered}}$$

I can think of at least three methods for finessing this problem. Measurements at a various distances, goniometry, and using very thin samples in which the contribution from scattering is negligible. The last two methods are described in the paper by Jacques, Alter, and Prahl which should be part of this distribution. The other method has not been published (as far as I know—since I thought it up and have not had the time to go to all the trouble to write it up and review the literature.)

I should probably mention that polarization can also be used to increase the collimated to scattered ratio. Unfortunately this only attenuates the diffuse contribution by a factor of two for thick samples (whose diffuse transmission will be unpolarized.)

If the light incident is perfectly collimated (no divergence) and that the scattered light is isotropically distributed over all angles. Then the amount of light detected at a distance d from the sample is

$$\begin{bmatrix} \text{Power} \\ \text{Measured} \\ [\text{W}] \end{bmatrix} = \begin{bmatrix} \text{Scattered} \\ \text{Intensity} \\ [\text{W sr}^{-1}] \end{bmatrix} \times \begin{bmatrix} \text{Solid Angle} \\ \text{of Detector} \\ [\text{sr}] \end{bmatrix} + \begin{bmatrix} \text{Collimated} \\ \text{Irradiance} \\ [\text{W cm}^{-2}] \end{bmatrix} \times \begin{bmatrix} \text{Area} \\ \text{of Detector} \\ [\text{cm}^2] \end{bmatrix}$$

If the scattered light emitted from the back of the sample is isotropic then the intensity for this light is a constant I_{scat} . The solid angle subtended by the detector located a distance d from the sample is given by

$$\text{detector solid angle} = \frac{\text{area of detector}}{\text{area of sphere with radius } d}$$

The average power per unit area is obtained by dividing the power measured by the area of the detector. The measured transmission through the sample is the ratio of the average power per unit area with and without the sample.

$$T_{\text{measured}} = \frac{\text{power per unit area with sample}}{\text{power per unit area without sample}}$$

d	d^{-2}	T	A	T	A
0.140	51	0.00288	2.54	0.00575	2.24
0.165	37	0.00234	2.63	0.00457	2.34
0.190	28	0.00195	2.71	0.00363	2.44
0.215	21	0.00151	2.82	0.00295	2.53
0.230	19	0.00123	2.91	0.00229	2.64

Table 7.1: Measurements made with a spectrophotometer and a piece of paper.

d	d^{-2}	paper		aorta	
		T	A	T	A
2.90	0.12	0.0000141	4.85	0.0000435	4.36
2.60	0.15	0.0000200	4.70	0.0000723	4.14
2.20	0.21	0.0000331	4.48	0.000145	3.84
1.90	0.28	0.0000447	4.35	0.000155	3.81
1.60	0.39	0.0001230	3.91	0.000263	3.58
1.30	0.59	0.0002460	3.61	0.000408	3.38
1.00	1.00	0.0004680	3.33	0.000629	3.20
0.48	4.30	0.0007080	3.15		
0.28	12.8	0.0012300	2.91	0.00175	2.75
0.08	156.	0.00260	2.60	0.00492	2.31

Table 7.2: Measurements made with a radiometer and a piece of paper.

or

$$T_{\text{measured}} = \frac{I_{\text{scat}}}{4\pi d^2 E_0} + \frac{E_{\text{coll}}}{E_0}$$

Notice that the measured transmission is independent of the area of the detector. The ratio of the collimated irradiance measured with and without the sample equals the collimated transmission so

$$T_{\text{measured}} = \frac{\text{constant}}{d^2} + T_{\text{coll}}$$

T_{coll} is the limit of as the separation between sample and detector becomes infinite. Thus, if T_{measured} is known for several distances between sample and detector, linear regression will yield both T_{coll} and the scattered intensity.

A major assumption in this development is that the scattered light exiting the sample is isotropic. This is true for optically thick samples but definitely not for optically thin ones [6].

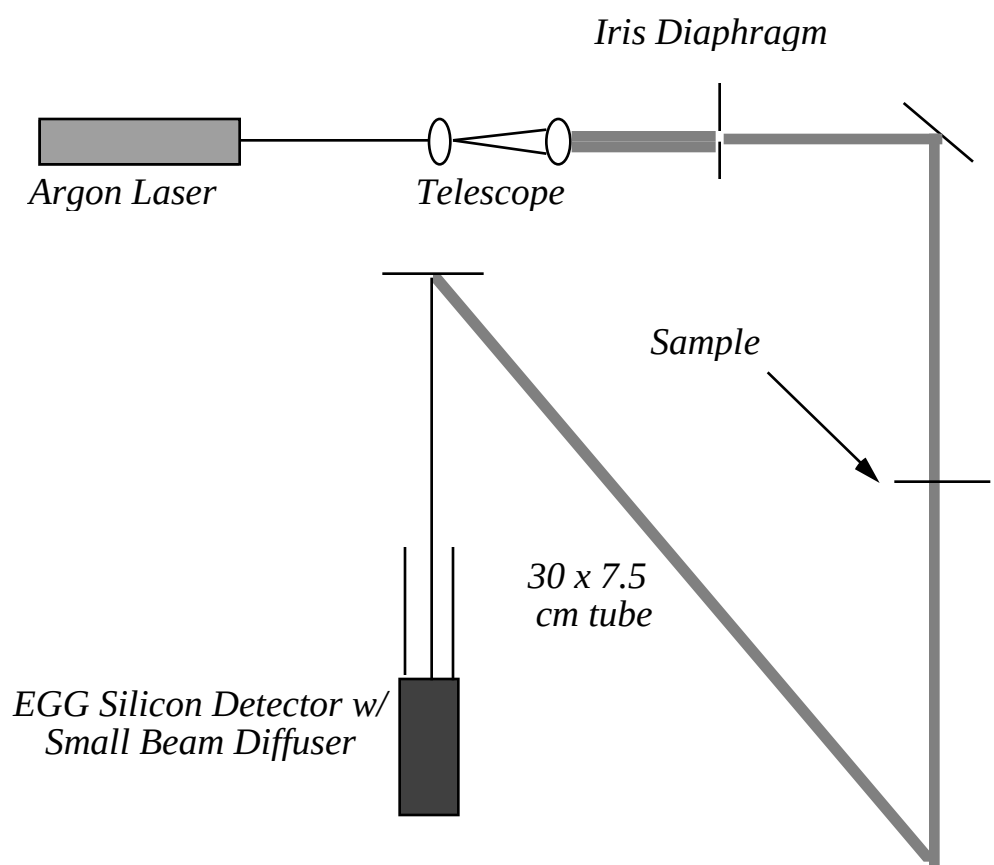


Figure 7.2: Diagrams showing how the reference measurements are made.

Chapter 8

Spheres Stuff

Now the theory of integrating spheres has been discussed many times by many people. I will add to the discussion once again

If the total surface area of a sphere (without holes) is A , then the normalized surface areas for the detector is $a_d = A_d/A$. The following indicates that the normalized area of the walls of the sphere

$$a_w = 1 - a_d - a_s - a_h$$

is one minus the areas of the detector, the sample, and all other holes in the sphere.

8.1 Signal when light hits the sphere wall first

Light enters the sphere with a power P . If the light hits the sphere wall first, then the amount of light reflected is

$$P^{(1)} = r_w P.$$

The reflected light is Lambertian and the amount of light falling on any particular area will be the ratio of that area to the total area. For example, the amount of light falling on the detector from the first reflection will be

$$P_d^{(1)} = (A_d/A)P^{(1)} = a_d P^{(1)} = a_d r_w P$$

The total amount of light falling on the walls of the sphere will be $a_w P^{(1)}$ and therefore the amount of light reflected inside the sphere will be the amount reflected by the walls plus that from the detector plus that from the sample

$$P^{(2)} = a_w r_w P^{(1)} + a_d r_d P^{(1)} + a_s r_s P^{(1)} = (a_w r_w + a_d r_d + a_s r_s) P^{(1)}$$

The amount of light falling on the detector from light that has been reflected twice by the sphere

$$P_d^{(2)} = a_d P^{(2)} = a_d a_w r_w P^{(1)} = a_d r_w (a_w r_w + a_d r_d + a_s r_s) P$$

The amount of light falling on the detector from the n th pass will be

$$P_d^{(n)} = a_d r_w (a_w r_w + a_d r_d + a_s r_s)^{n-1} P$$

The total amount of light falling on the detector for light incident first on the wall of the sphere is

$$P_d = \frac{a_d r_w}{1 - a_w r_w - a_d r_d - a_s r_s} P$$

obtained by summing the infinite series. The factor

$$\eta_s = \frac{a_d}{1 - a_w r_w - a_d r_d - a_s r_s}$$

occurs is sometimes called the sphere efficiency factor. Therefore,

$$P_d = \eta_s r_w P$$

Now if the sample is illuminated first with collimated light and the sample reflects the light both specularly r_s^{cc} and diffusely r_s^c , then there will be $r_w r_s^{cc} P$ diffuse light created by the specular reflection upon reaching the sphere wall, and $r_s^c P$ through diffuse reflection. The amount of light detected will be

$$P_s = \eta_s (r_w r_s^{cc} + r_s^c) P$$

If the sample is replaced by a reflection standard then

$$P_{\text{std}} = \eta_{\text{std}} (r_w r_{\text{std}}^{cc} + r_{\text{std}}^c) P$$

The ratio is

$$\frac{P_s}{P_{\text{std}}} = \frac{r_w r_s^{cc} + r_s^c}{r_w r_{\text{std}}^{cc} + r_{\text{std}}^c} \frac{\eta_s}{\eta_{\text{std}}} = \frac{r_w r_s^{cc} + r_s^c}{r_w r_{\text{std}}^{cc} + r_{\text{std}}^c} \left[1 - \frac{(r_{\text{std}} - r_s) a_s}{1 - a_w r_w - a_d r_d - a_s r_s} \right]$$

Typically a Spectralon reflection standard will be used. The amount of light reflected from these sample is nearly equal for collimated and diffuse irradiance. This means $r_{\text{std}} = r_{\text{std}}^c$ and $r_{\text{std}}^{cc} = 0$, and so

$$\frac{P_s}{P_{\text{std}}} r_{\text{std}} = (r_s^c + r_s^{cc} r_w) \left[1 - \frac{(r_{\text{std}} - r_s) a_s}{1 - a_w r_w - a_d r_d - a_s r_s} \right]$$

Now if the sample is positioned so that the specular reflection leaves the sphere without striking the inner wall

$$\frac{P_s}{P_{\text{std}}} r_{\text{std}} = r_s^c \left[1 - \frac{(r_{\text{std}} - r_s) a_s}{1 - a_w r_w - a_d r_d - a_s r_s} \right]$$

The error introduced by the spheres is proportional to the difference between the reflectance of the calibration standard and the reflectance of the sample,

$$\text{error} = \frac{(r_{\text{std}} - r_s) a_s}{1 - a_w r_w - a_d r_d - a_s r_s}$$

8.2 The reflectance of the sphere wall

The voltage is proportional to the power incident on the detector (with proportionality constant k , therefore

$$V_{\text{empty}} = \frac{a_d r_w k P}{1 - a_w r_w - a_d r_d} \quad \text{and} \quad V_{\text{std}} = \frac{a_d r_w k P}{1 - a_w r_w - a_d r_d - a_s r_{\text{std}}}$$

The constant of proportionality is unknown. One expression for the reflectance of the sphere wall is

$$r_w = 1 - a_d r_d - \frac{a_s r_s}{1 - V_{\text{nothing}}/V_{\text{reference}}}$$

Now if the error in the voltage is the same for both measurements ΔV and all the port sizes are known with good accuracy, then the error in r_w will be approximately

$$\Delta r_w \approx 2(1 - a_d r_d - r_w) \frac{\Delta V}{V_{\text{reference}} - V_{\text{nothing}}}$$

8.3 The sphere parameter

Is all this hullabaloo really necessary? Well, consider the usual way that reflection coefficients are measured. Assume collimated irradiance.

In this case, the voltage detected is

$$V = \frac{a_d r_w r_s^{cc} + a_d r_s^c}{1 - a_w r_w - a_s r_s - a_d r_d} P$$

This can be rewritten

$$V = \frac{a_d r_w r_s^{cc} + a_d r_s^c}{1 - a_s r_s x} x P$$

where

$$x = \frac{1}{1 - a_w r_w - a_d r_d}$$

8.4 Calibration

This section will be incoherent to nearly everyone. The method has not been implemented in the code yet. No time. But it is included here for completeness.

The basic idea is that the corrections necessary for any double sphere measurement can be calculated by using the integrating sphere formulas. This means that five measurements are necessary for determining three optical properties: $V_{1\%}$, $V_{2\%}$, $V_{3\%}$, $V_{10\%}$, $V_{20\%}$. The % just indicates that all measurements are relative to a reference measurement in which any signal that is not proportional to the incident irradiance has been removed, for example,

$$V_{\%} = \frac{V_1 - V_{\text{dark}}}{V_{\text{ref}} - V_{\text{dark}}}.$$

The idea is to correct $V_{1\%}$ and $V_{2\%}$ by

$$V_{1\%}^{\text{corrected}} = V_{1\%} - F_1 V_{10\%} - F_2 V_{20\%}$$

and

$$V_{2\%}^{\text{corrected}} = V_{2\%} - F_3 V_{10\%} - F_4 V_{20\%}$$

Stray light is proportional to the incident irradiance and arises from light which does not escape the spheres when no sample is present. The appropriate correction for any particular measurement depends on the sample. The correction necessary for stray light in the reflection sphere is different from the correction for stray light in the transmission sphere. Fortunately the stray light sources can be measured independently. Baseline measurements with both spheres in place are not very useful because they include the stray light effects from both spheres. It is better to make careful measurements with single spheres since the contribution from each sphere can be easily separated.

Let $V_{10\%}$ be the detected signal in the reflectance sphere with no sample in the port and no transmission sphere present. Let $V_{20\%}$ be the detected signal in the transmission sphere with no sample and the reflection sphere absent. In both cases the spheres should be in exactly the same place relative to the source as they will be when the sample is measured.

$V_{10\%}$ and $V_{20\%}$ should be thought of as two diffuse sources for stray light. The contribution of $V_{10\%}$ to the signal in each sphere in the presence of a sample can be calculated by assuming that the sample is illuminated by a diffuse source with power $V_{10\%}$. The correction for the reflectance sphere should be something like the double sphere formula for diffuse irradiance. However, since $V_{10\%}$ is the light incident on the detector it no longer must be corrected for that first reflection from the sphere wall. Therefore both correction factors are divided by b_{1r}

$$F_1 = \frac{(1 - R_d \frac{A_s}{A_r})(1 - b_{2t} R_d)}{(1 - b_{2r} R_d)(1 - b_{2t} R_d) - b_{2r} b_{2t} T_d^2}$$

The correction for the transmission sphere should be

$$F_3 = \frac{1}{b_{1r}} \frac{b_{1t} b_{2r} m_r \alpha_t T_d}{(1 - b_{2r} R_d)(1 - b_{2t} R_d) - b_{2r} b_{2t} T_d^2}$$

The corrections arising from $V_{20\%}$ can be worked out in a similar fashion. However the quantity $V_{20\%}$ must now be multiplied by T_c and divided by b_{1t} . Moreover, the r 's and t 's should be swapped. The correction for the transmission sphere is

$$F_4 = \frac{T_c}{b_{1t}} \frac{(1 - R_d \frac{A_s}{A_t})(1 - b_{2r} R_d)}{(1 - b_{2r} R_d)(1 - b_{2t} R_d) - b_{2r} b_{2t} T_d^2}$$

The correction for the reflection sphere should be

$$F_2 = \frac{T_c}{b_{1t}} \frac{b_{1r} b_{2t} m_t \alpha_r T_d}{(1 - b_{2r} R_d)(1 - b_{2t} R_d) - b_{2r} b_{2t} T_d^2}$$

Chapter 9

Questions

9.1 What is this business about V%?

The integers following the V refer to the place where the measurement took place. The first measurement $V_1\%$ always refers to a reflection measurement. This may include specular reflection or not. The second $V_2\%$ is a transmission measurement which may or may not include unscattered transmission. The third $V_3\%$ is a collimated or unscattered light measurement.

The percent sign indicates that this quantity has been normalized to some baseline. **All values lie between 0 and 1.** They are not true percentages and the notation leaves something to be desired, but you get what you pay for. The notation $V\%$ refers to the quantity

$$V\% = \frac{V_{\text{measure}} - V_{\text{nothing}}}{V_{\text{reference}} - V_{\text{nothing}}}$$

Where V is the voltage measured of the light detector. If an integrating sphere is used and $V_{\text{reference}}$ refers to 100% reflection then $V\%$ is a first order approximation to the reflectance of the sample.

9.2 How are the sphere coefficients determined?

I could direct your attention to the Pickering paper. Unfortunately John's description of the process is cryptic at best. So I won't. In any case I begin by noting that all measurements should be normalized by the total power from your light source. In particular, the normalized percentage $V_1\%$ or fraction as it more properly should be called is

$$V_1\% = \frac{P_d}{P} = \frac{V_1 - V_0}{V_{\text{ref}} - V_{\text{ref},0}}$$

where the stuff in the denominator is a measurement of the total power incident on the sample. All the reference measurements are made with a single sphere

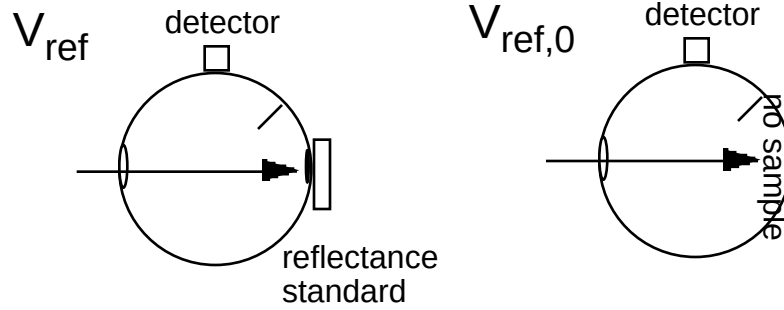


Figure 9.1: Diagrams showing how the reference measurements are made.

(figure 9.1). These are just measurements to normalize the measured power to the total power. One should be careful to ensure that $V_{\text{ref},0}$ is nearly the same as the measurement when the beam is blocked.

To determine the sphere coefficient b_1 , Pickering observes that (actually van Wieringen) that for diffuse irradiance with no sample present

$$b_1 = V_1 \% = \frac{V_1 - V_0}{V_{\text{ref}} - V_{\text{ref},0}}$$

Reading Pickering *et al.* carefully, we find that V_1 is the signal measured when collimated light initially hits the sphere wall (making diffuse light) *with no sample present* (because $R_d \equiv 0$ in equation 1). The definition of V_0 is tricky. If we look at the text in Pickering *et al.* (and not in the table 1) then V_0 is a background signal that depends on the noise in the detection system and on the presence of stray light. The only measurement that makes sense in this case is

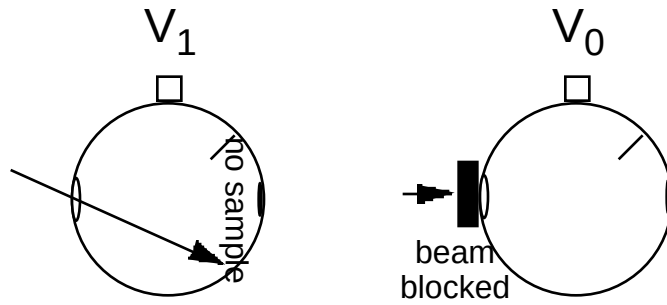
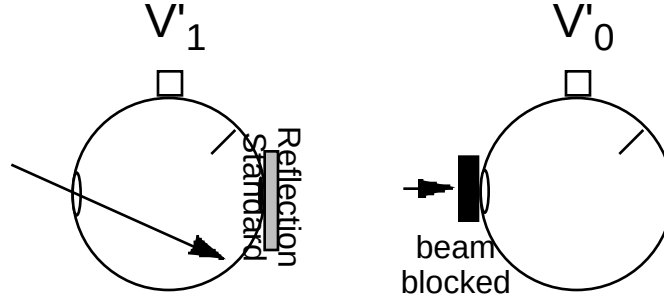


Figure 9.2: The measurements necessary to determine b_1

when the beam is blocked before it enters the sphere. These experiments are diagrammed in Figure 9.2. The V_0 as defined by Pickering's Table 1 is exactly the same as the V_1 measurement just described.

Figure 9.3: The measurements necessary to determine b_2

To determine b_2 , we place a reflection standard in the port. Again, because the illumination is diffuse, the baseline measurement must be made with the beam blocked (Figure 9.3). Now if

$$V'_1\% = \frac{V'_1 - V'_0}{V_{\text{ref}} - V_{\text{ref},0}}$$

Then Pickering in equation (20) suggests that b_2 is

$$b_2 = \frac{b_1 - V'_1\%}{V'_1\% R_{ds}} \quad \text{wrong}$$

where R_{ds} is the known diffuse reflectance of the standard. It is probably worthwhile to note that the reference measurements for b_2 cancel each other. In fact we find that

$$b_2 = \frac{V_1 - V'_1}{R_{ds}(V'_1 - V_0)} \quad \text{wrong}$$

To interpret this physically lets make the assumption that the reflection standard is 100% reflecting. Furthermore, let's assume that the background measurement V'_0 is negligible. In this case

$$b_2 = \frac{V_1}{V'_1} - 1 \quad \text{wrong}$$

or that b_2 is always negative. Oops! This must be wrong. Going back and checking equation (20) in the Pickering paper, I find that it should actually read,

$$b_2 = \frac{V'_1\% - b_1 + b_1 R_{ds} a_s}{V'_1\% R_{ds}}$$

If we neglect the effect of the baffle, then

$$b_2 = 1 - \frac{V_1}{V'_1}$$

Or that b_2 is the fraction of diffuse light that is affected by the presence of a sample. For example, if V_1/V_1' is 0.9 (likely for small port sizes) then the maximum change that a sample can have is only 10% of the measurement. This is why diffuse light measurements are frowned upon.

The coefficient a_w is best determined geometrically.

9.3 Yes, but how do I really determine do it?

Let's look at the definitions for b_1 and b_2 .

$$b_1 = \frac{r_w a_d}{1 - r_w a_w} \quad b_2 = \frac{a_s}{1 - r_w a_w}$$

The term $1 - r_w a_w$ can be small. This means that the quantities b_1 and b_2 are the ratio of two small numbers. If either a_w (the ratio of the surface area of the sphere with holes to the area without holes) is not close to unity or the reflectivity of the sphere is not nearly one, then the effect will be mitigated. For example in the limiting case of a large sphere with perfect reflectivity $r_w = 1$ (and only sample and detector holes — no hole for the entering laser beam) we get

$$b_1 = \frac{a_d}{a_d + a_s} \quad b_2 = \frac{a_s}{a_d + a_s}$$

Assuming that the sample size is larger than the detector port

$$b_1 = \frac{\frac{a_d}{a_s}}{1 + \frac{a_d}{a_s}} \sim \frac{a_d}{a_s} \quad b_2 = \frac{1}{1 + \frac{a_d}{a_s}} \sim 1 - \frac{a_d}{a_s}$$

As a check, note that $b_1/b_2 = r_w a_d/a_s$ and the above expressions are consistent.

However, this little excursion implicitly assumes that

$$1 - r_w \ll a_s + a_d$$

which may be the case for small spheres. For example a six inch sphere with one-inch ports and a nominal value of 0.98 for r_w gives

$$0.02 \ll \frac{\pi 0.25^2 + \pi 0.5^2}{4\pi \cdot 3^2} = 0.06$$

However, for a monster twenty inch sphere with a two inch sample port and a quarter inch detector port gives

$$0.02 \ll \frac{\pi 2^2}{4\pi 10^2} = 0.01$$

Now if we take the opposite tack and assume that $a_w \approx 1$ then we don't get a whole lot of simplification so we'll just try some numbers out. Let's assume the sphere is twenty inches, the sample port is two inches, the detector port is a quarter inch, that $\alpha = 0.9837$ and $r_w = 0.9856$ then

$$b_1 \approx \frac{0.125^2}{400} \frac{0.9856}{1 - 0.9856} = 0.0027 \quad b_2 \approx \frac{1}{400} \frac{1}{1 - 0.9856} = 0.17$$

substituting numbers

$$b_1 \approx \frac{0.125^2}{4 \cdot 10^2} \frac{0.98}{0.02} \sim 1/512 \quad b_2 \approx \frac{1}{4 \cdot 10^2} \frac{1}{0.02} = 1/8$$

and $b_1/b_2 = 1/64$.

Yes, but how should I determine b_1 and b_2 ? Well, using the method described above (ignoring the baffle) one would dutifully measure

$$b_1 = \frac{V_1 - V_0}{V_{\text{ref}} - V_{\text{ref},0}} \quad b_2 = \frac{V'_1 - V_1}{V'_1}$$

Now the difference between V'_1 and V_1 will most likely be quite small. Taking the difference of two large numbers is not so good experimentally.

Instead, let's just depend on one measurement of the reflectance of the integrating sphere. To determine r_w , measure $V_1\%$

$$b_1 = a_d \frac{r_w}{1 - r_w a_w} = V_1\%$$

solving for r_w gives

$$r_w = \left(a_w + \frac{a_d}{V_1\%} \right)^{-1}$$

then

$$r_w = \frac{1}{1 - a_s - a_h + a_d(1 - V_1\%)/V_1\%}$$

which is a very nice expression because the one geometric quantity that is most poorly known is the detector area. Anyway, I see now that the best way to have people input the sphere coefficients is by reporting the ratio of the port sizes a_s , a_h , a_d and r_w . This will be a reality in version 1.2 of the IAD program.

9.4 How should single sphere measurements be made?

So what are the other measurements that must be made? Well, clearly we must determine $V_1\%$ and $V_2\%$. First of all, if only a single sphere is being used then the measurements are relatively straightforward. The reference measurements will be exactly the same as in Figure 9.1. The other measurements should be done as in figure 9.5 and 9.4.

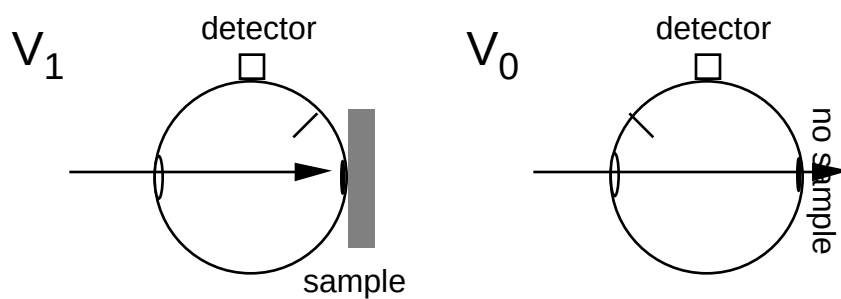


Figure 9.4: The measurements to make a $V_1\%$ measurement.

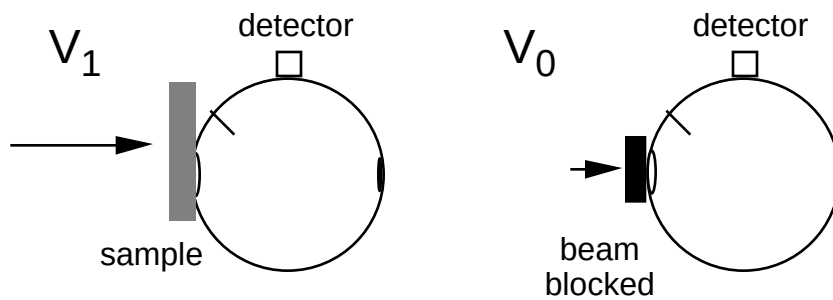


Figure 9.5: The measurements to make a $V_2\%$ measurement.

Chapter 10

Implementation

This section must be rewritten to reflect that the program is in C and not Pascal. I just have not had the time to do anything about it. Sorry.

You probably don't have the source code for this program and should probably ignore this section.

It should not be necessary to change anything. But of course everyone will have slightly different needs. This section will give an overview of the program and sketch how the program may be customized.

The whole program basically revolves around a call to the procedure

```
procedure RT_inverse(m: measure_type; var r:result_type);
```

You the excitable programmer will spend all energies setting the measurement specified by the record `m` up for this procedure call. The result will be returned in the record `r`. There are a bunch of routines for fiddling with the measurement record in the `ad_inv_io` unit (*adding-doubling inverse input/output* unit).

This is a list of the fundamental routines for input/output operations.

```
{Routines to initialize data structures}
function default_measure: measure_type;
function default_slab: AD_slab_type;

{routines to input from the keyboard}
function input_measure: measure_type;
procedure input_voltages (var m: measure_type; var done: boolean);

{Routines to read from a data file}
procedure read_measure (var t: text; var m: measure_type);
procedure read_voltages (var t: text; var m: measure_type; var done: boolean);

{Routine to write to a file}
procedure write_measure (var t: text; m: measure_type);
```

These are used a couple of sections farther down.

10.1 The Measurement Data Structure

As you can see below, there are some thirty-six parameters that must be set correctly in the measurement record. The easiest way to set them all is to call the procedure `default_measure`. Then set the few values that you need to change. Another equally good way is to create a file with a header as described above and call the routine `read_measure`.

```

program example;
const
  max_streams = 16;
  mp = 3;
  np = 2;
type
  phase_function_type = (Henyey-Greenstein, Rayleigh, Mie, Isotropic,
                        Diffusion, modified_henyey-greenstein);
  AD_slab_type = record
    a: real;           {albedo of the slab}
    b: real;           {optical thickness of the slab}
    phase_function: phase_function_type;
    g: real;           {ave cos of the phase function}
    extra: real;       {second phase fn parameter if needed}
    n_slab: real;      {index of refraction of the slab}
    n_top_slide: real; {index of refr. of a glass slide on slab}
    n_bottom_slide: real; {index of refr. of slide beneath slab}
  end ;

  simplex = array[1..mp, 1..np] of real;
  search_type = (find_ab, find_ag, find_a, auto);

  measure_type = record
    slab: AD_slab_type;      {sample information}
    illumination: (collimated, diffuse); {simple enough}
    num_spheres: 0..2;       {number of integrating spheres used}
    num_measures: 1..3;      {number of measurements made}
    v1, v2, v3: real;        {refl, trans, and coll refl}
    b11,b12,b13: real;       {sphere coefficients for sphere 1}
    b21,b22,b23: real;       {sphere coefficients for sphere 2}
    mfactor: real;           {refl. factor for sphere 1 }
    sphere_with_rs: boolean; {if sphere 1 collects spec refl}
    {calculation information}
    nfluxes: 2..max_streams; {number of fluxes...must be even}
    search: search_type;     {type of inverse search used}
    tolerance: real;         {ending tolerance}
    default_g: real;         {value of g to use when V3=0}
    thickness: real;         {physical thickness of sample in mm}
  end ;

```

```

    use_last_simplex: boolean;
    last_p: simplex;
end;

```

10.2 Result Data Structure

The data structure for the result data type is given by

```

type
  result_type = record
    a: real;
    b: real;
    g: real;
    mu_a: real;
    mu_s: real;
    found: boolean;
    tol: real;
    iterations: integer;
    final_p: simplex;
  end;

```

The answer is returned in the result record. If `r.found` is true then the optical properties were found. If it is false then there was some trouble and the optical properties returned must be regarded with *great suspicion*. The values for `s` and `a` are given by `r.mu_s` and `r.mu_a` respectively. However, these are calculated based on the depth passed in the measurement record. Moreover, the units are the inverse of those. An interesting value returned is `r.iterations` which not surprisingly counts the number of simplex iterations. Multiply this by about four and you will obtain the number of times the reflection and transmission was calculated.

10.3 Writing Your Own Program

Example 1

Suppose you have a data file which has values for the reflection and transmission as a function of wavelength. Unfortunately the first column is the wavelength. Moreover, you want to optimize to program so that the optical properties for the last point are used as new starting value.

```

program spectrum;
uses
  ad_globals, ad_inv_io, ad_inv_public;
var
  m: measurement_type;

```

```
r: result_type;
filename: string;
infile: text;
lambda: real;
begin
  filename:=oldfilename('');
  if filename<>' ' then
    begin
      reset(infile,filename);
      read_measure(infile,m);
      while not eof(infile) do
        begin
          readln(infile, lambda, m.v1, m.v2);
          inverse_rt(m,r);
          write(lambda:10:5, r.a:10:5, r.b:10:5, r.g:10:5);
          if r.found then writeln(' !') else writeln(' ?');
          m.use_last_simplex := r.found;
        end;
      end;
    end;
  end.
```

Piece of cake right! By the way, this program will die if there is an extra return at the end of the data file.

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